

```
# uEnv.txt

uname_r=5.10.0-bone-rt

bootargs=console=ttYO0,115200n8 root=/dev/mmcblk0p2 rootfstype=ext4 rootwait

bootfile=zImage

fdtfile=am335x-boneblack.dtb

mmccroot=/dev/mmcblk0p2
```

U-Boot SPL 2020.10 (Oct 01 2020 - 12:34:56)

Trying to boot from MMC1

U-Boot 2020.10 (Oct 01 2020 - 12:34:56 -0500)

CPU : AM335X-GP rev 2.1

Model: BeagleBone Black

Board: TI AM335x BeagleBone Black

DRAM: 512 MiB

MMC: OMAP SD/MMC: 0, OMAP SD/MMC: 1

In: serial

Out: serial

Err: serial

Net: eth0: ethernet@4a100000

Hit any key to stop autoboot: 0

reading uEnv.txt

Loading environment from mmc 0:1/uEnv.txt

404 bytes read in 3 ms (131.9 KiB/s)

Loaded environment from /boot/uEnv.txt

Importing environment from mmc ... OK

MMC: no card present

mmc0: SD/MMC found

mmc1: SD/MMC found

reading zImage

1100000 bytes read in 120 ms (8.8 MiB/s)

reading am335x-boneblack.dtb

15000 bytes read in 4 ms (3.7 MiB/s)

Booting kernel from Legacy Image at 82000000 ...

Image Name: Linux-5.10.0-bone-rt

Image Type: ARM Linux Kernel Image (uncompressed)

Data Size: 1100000 Bytes = 1.0 MiB

Load Address: 80008000

Entry Point: 80008000

Verifying Checksum ... OK

Loading Kernel Image ... OK

Loading Device Tree to 83fff000, end 84000000 ... OK

Starting kernel ...

[0.000000] Booting Linux on physical CPU 0x0

[0.000000] Linux version 5.10.0-bone-rt (root@buildhost) (gcc version 10.2.1 20201203) #1 SMP PREEMPT RT

[0.000000] CPU: AM335X-GP rev 2.1

[0.000000] Machine: BeagleBone Black

[0.000000] Memory policy: Data cache writeback

[0.000000] efi: Getting EFI parameters from FDT:

[0.000000] efi: EFI not found.

[0.000000] OF: fdt: Machine model: TI AM335x BeagleBone Black

[0.000000] cma: Reserved 16 MiB at 0x80000000

[0.000000] Zone ranges:

[0.000000] Normal [mem 0x0000000080000000-0x0000000081fffff]

[0.000000] HighMem empty

[0.000000] Movable zone start for each node

```
[ 0.000000] Early memory node ranges
[ 0.000000] node 0: [mem 0x0000000080000000-0x0000000081ffffff]
[ 0.000000] Initmem setup node 0 [mem 0x0000000080000000-0x0000000081ffffff]
[ 0.000000] percpu: Embedded 14 pages/cpu s40240 r8192 d22816 u57344
[ 0.000000] Built 1 zonelists in Node order, mobility grouping on. Total pages: 130560
[ 0.000000] Kernel command line: console=ttyO0,115200n8 root=/dev/mmcblk0p2 rootfstype=ext4 rootwait
[ 0.000000] Dentry cache hash table entries: 65536 (order: 6, 262144 bytes)
[ 0.000000] Inode-cache hash table entries: 32768 (order: 5, 131072 bytes)
[ 0.000000] Memory: 502640K/524288K available (12345K kernel code, 2345K rwddata, 5120K rodata,
2048K init, 1234K bss, 16384K reserved, 16384K cma-reserved)
[ 0.000000] Virtual kernel memory layout:
[ 0.000000] vector : 0xffff0000 - 0xffff1000 ( 4 kB)
[ 0.000000] fixmap : 0xffc00000 - 0xfff00000 (3072 kB)
[ 0.000000] vmalloc : 0x90000000 - 0xff800000 (1792 MB)
[ 0.000000] lowmem : 0x80000000 - 0x88000000 (128 MB)
[ 0.000000] modules : 0x7f000000 - 0x80000000 ( 16 MB)
[ 0.000000] CPU: Testing write buffer coherency: ok
[ 0.000000] CPU0: AM335X-GP rev 2.1 (ARMv7)
[ 0.000000] CPU: All CPU(s) started in SVC mode.
[ 0.000000] alternatives: patching kernel code
[ 0.000000] Detected VIPT I-cache on CPU0
[ 0.000000] Built 1 zonelists in Node order, mobility grouping on. Total pages: 130560
[ 0.000000] Kernel command line: console=ttyO0,115200n8 root=/dev/mmcblk0p2 rootfstype=ext4 rootwait
[ 0.000000] bootconsole [earlycon0] enabled
[ 0.000000] printk: console [ttyO0] enabled
[ 0.500000] omap_voltage_late_init: voltage driver initialized
[ 0.600000] omap_hwmod: hwmod: timer1 already enabled
[ 0.610000] omap_hwmod: hwmod: timer2 already enabled
[ 0.620000] omap_hwmod: hwmod: timer3 already enabled
[ 0.630000] omap_hwmod: hwmod: timer4 already enabled
[ 0.640000] Serial: AM335X UART driver
[ 0.650000] serial 48022000.serial: ttyO0 at MMIO 0x48022000 (irq = 72, base_baud = 3000000) is a
NS16550
```

[1.000000] mmc0: host does not support reading read-only switch, assuming write-enable

[1.010000] mmc0: new high speed SDHC card at address 1234

[1.020000] mmcblk0: mmc0:0001 00000 3.7 GiB

[1.030000] mmcblk0: p1 p2

[1.100000] VFS: Mounted root (ext4 filesystem) on device 179:2.

[1.120000] devtmpfs: mounted

[1.150000] Freeing unused kernel memory: 2048K

[1.200000] Run /sbin/init as init process

[1.250000] usbcore: registered new interface driver usbfs

[1.260000] usbcore: registered new interface driver hub

[1.270000] usbcore: registered new device driver usb

[1.300000] random: fast init done

[1.350000] ALSA sound initialized

[1.400000] mmc1: new SDHC card at address 1111

[1.500000] gpio-keys: registered

[1.600000] usb 1-1: new high-speed USB device number 2 using ehci-omap

[2.000000] EXT4-fs (mmcblk0p2): mounted filesystem with ordered data mode

[2.100000] Bone-capemgr: loaded

[2.150000] Bone-capemgr: slot #0: Generic Cape loaded

[2.200000] bone-capemgr: initialized OK

[2.250000] IPv6: ADDRCONF(NETDEV_UP): eth0: link is not ready

[2.300000] usb 1-1: new high-speed USB device number 3 using ehci-omap

[3.000000] eth0: link up, 100Mbps, full-duplex

[3.050000] random: crng init done

[3.100000] init: version 247.3-7ubuntu1

[3.150000] init: entering runlevel: 2

[3.200000] Welcome to Debian 10 (buster)!